



New Heights Old Traditions

# **TAMILNADU COLLEGE OF ENGINEERING**

**Accredited by NAAC with Grade 'A'**

**(Approved by AICTE & Affiliated to Anna University, Chennai)**



**Department of Computer Science and Engineering**

## **Project Final Review**

**DATE : 09.04. 2026**

# Session Based – Cloud Integrated File Transferring Ecosystem

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# Introduction

In the modern digital era, secure and efficient sharing of confidential documents has become a critical requirement for individuals, organizations, and enterprises. Traditional file-sharing methods often lack sufficient security measures, making sensitive information vulnerable to unauthorized access, data breaches, and cyber-attacks. To address these challenges, the Session Integrated Confidential Document Sharing Application is designed to provide a secure, reliable, and user-friendly platform for sharing sensitive documents. The system integrates session-based authentication and encryption techniques to ensure that only authorized users can access shared files within a limited time frame. This application enhances data confidentiality, integrity, and secure transmission, making it suitable for professional, academic, and business environments.

# Abstract

The Session Integrated Confidential Document Sharing Application is a secure file-sharing platform that enables users to transfer sensitive documents safely using session-based authentication and encryption techniques. The system establishes a secure session between the sender and receiver, ensuring that documents are accessible only during the active session. Once the session expires, the shared files become inaccessible, thereby preventing unauthorized reuse or leakage. The application incorporates features such as encrypted data transfer, real-time device discovery, access control mechanisms, and secure storage handling. By combining these technologies, the system provides a robust solution for confidential document sharing, reducing security risks and improving trust in digital communication systems.

# Literature Survey

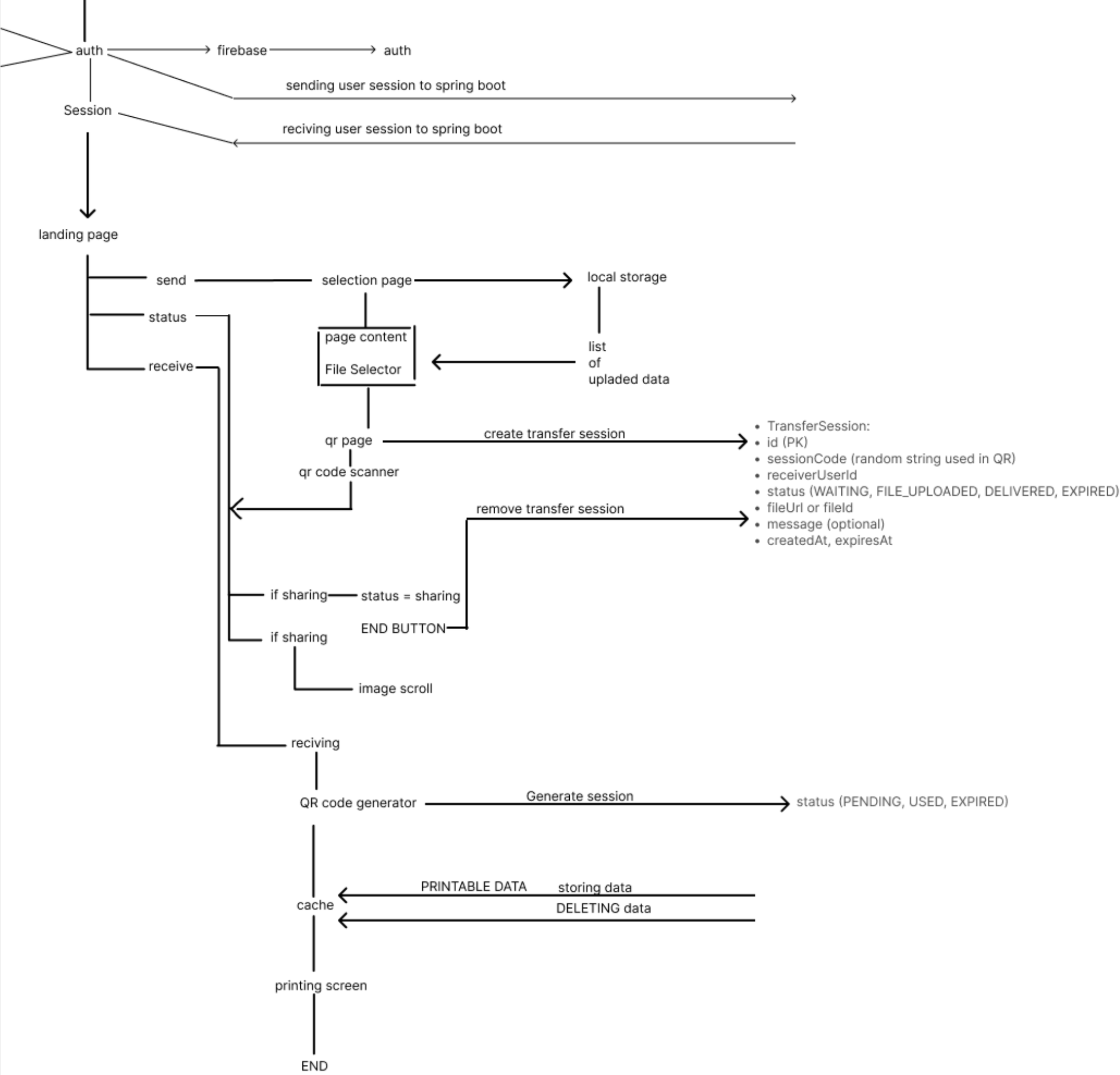
| Author & Year                | IEEE Paper Title                                             | Methodology / Tech                                                                           |
|------------------------------|--------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| Zhang, Y. & Lee, W. (2019)   | Secure Peer-to-Peer File Sharing Using End-to-End Encryption | End-to-End AES Encryption<br>RSA Key Exchange<br>Peer-to-Peer Communication Model            |
| Kumar, S. & Patel, R. (2020) | Session-Based Authentication for Secure Data Transmission    | Session Tokens<br>Time-Limited Access Control<br>HTTPS Secure Communication                  |
| Chen, L. et al. (2021)       | Secure Mobile File Sharing Using WiFi Direct                 | WiFi Direct Peer Discovery<br>Encrypted Socket Communication<br>Android-Based Implementation |



# Proposed Method

- Session based file transfer
- Cloud integration
- User authentication
- Data set shared via QR
- Session maintained by multilevel security

# Flow Chart

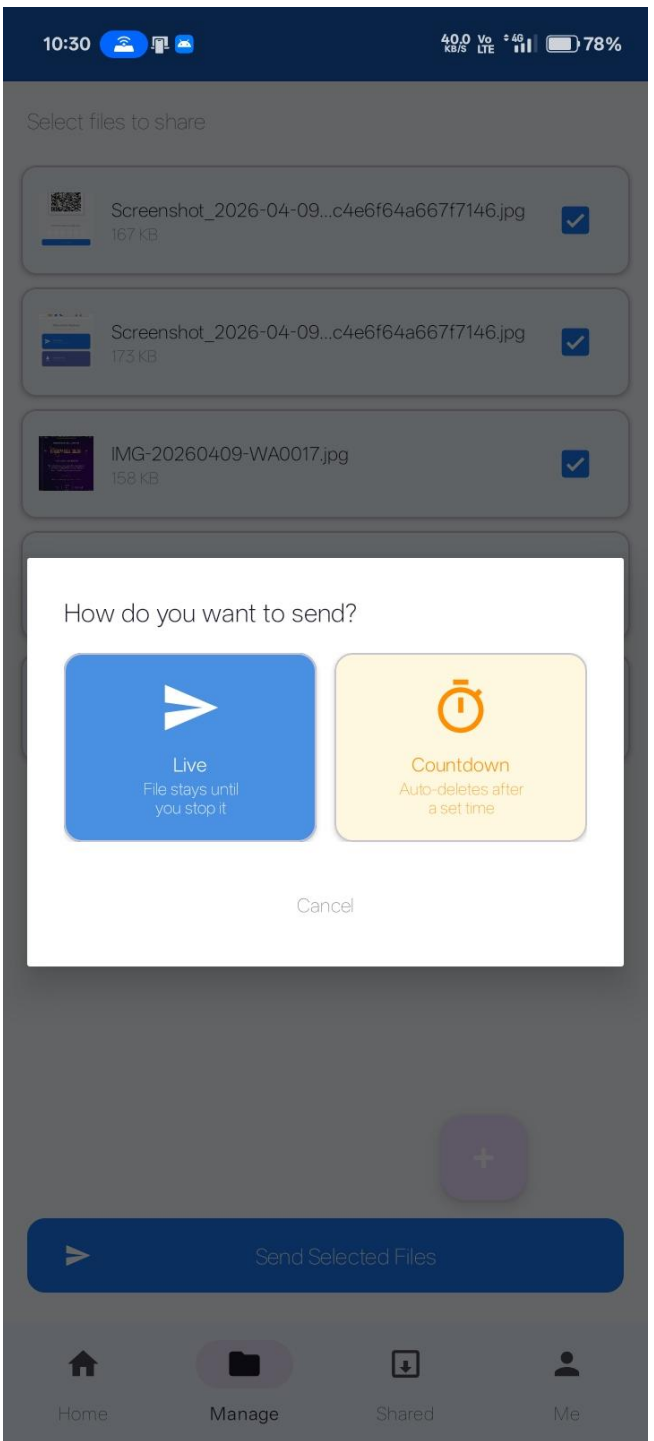
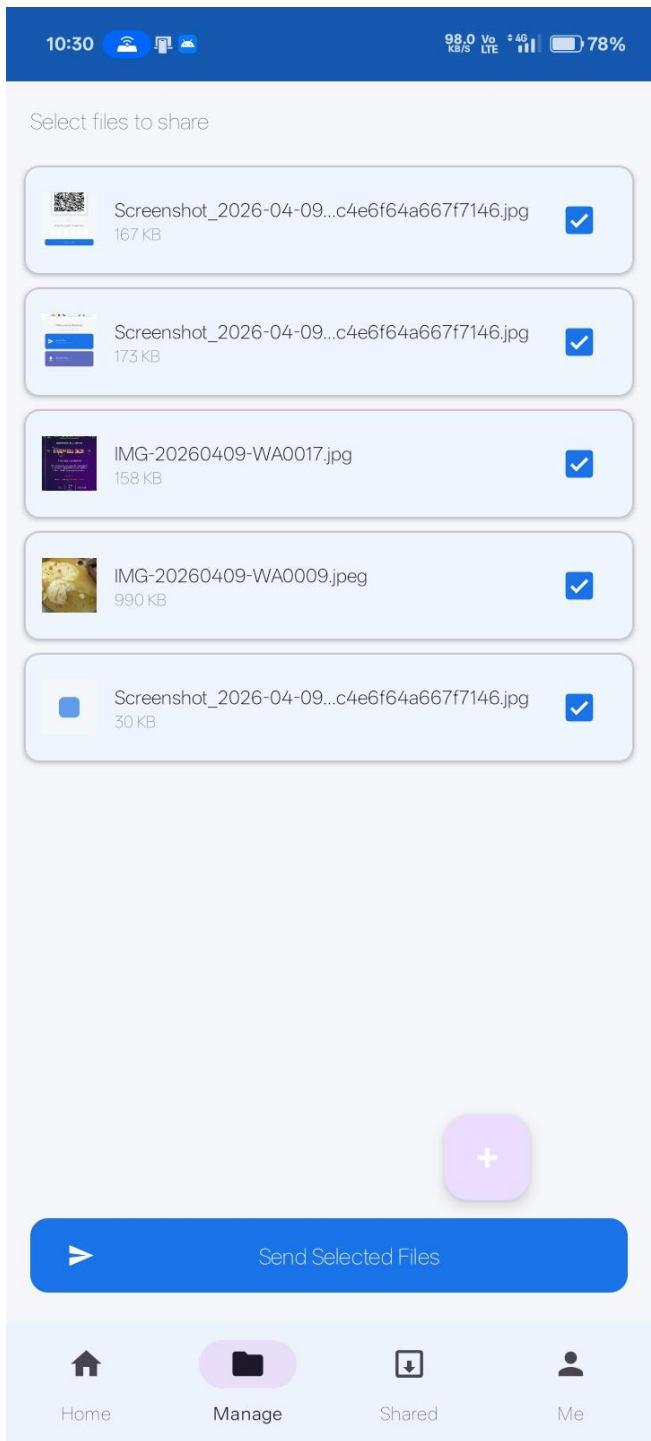
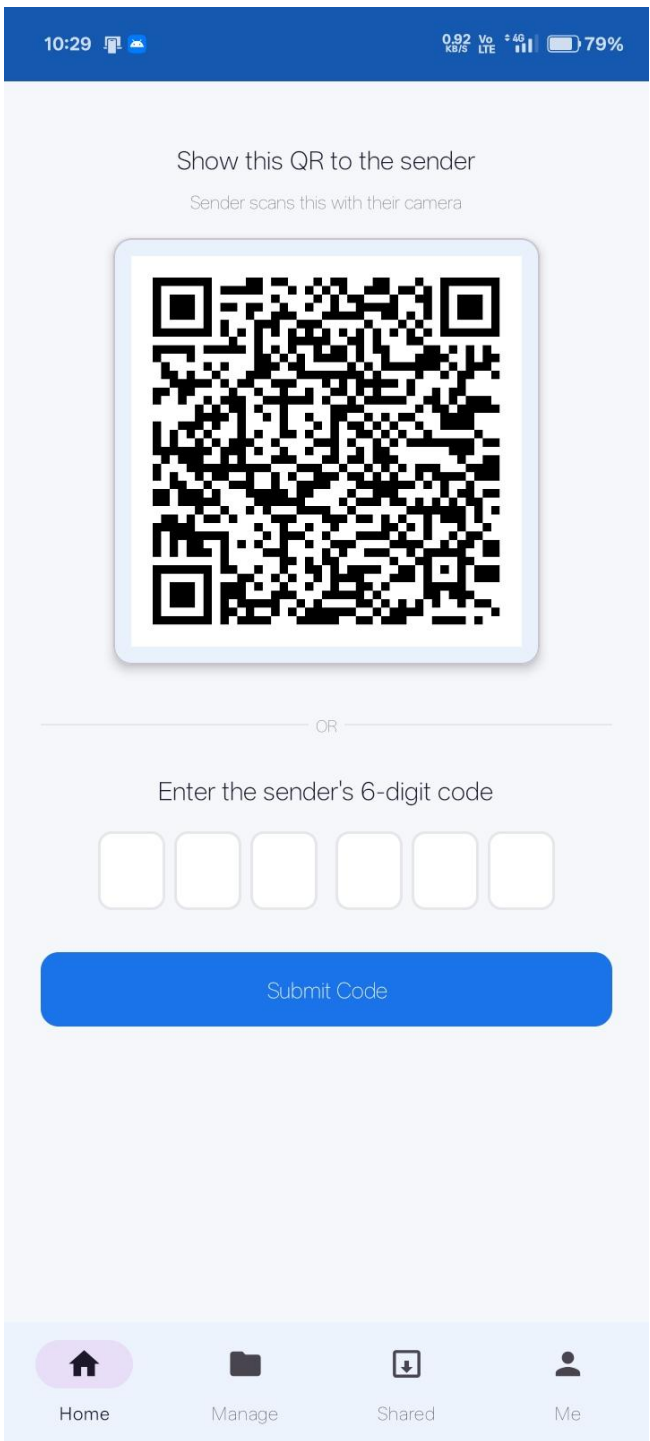
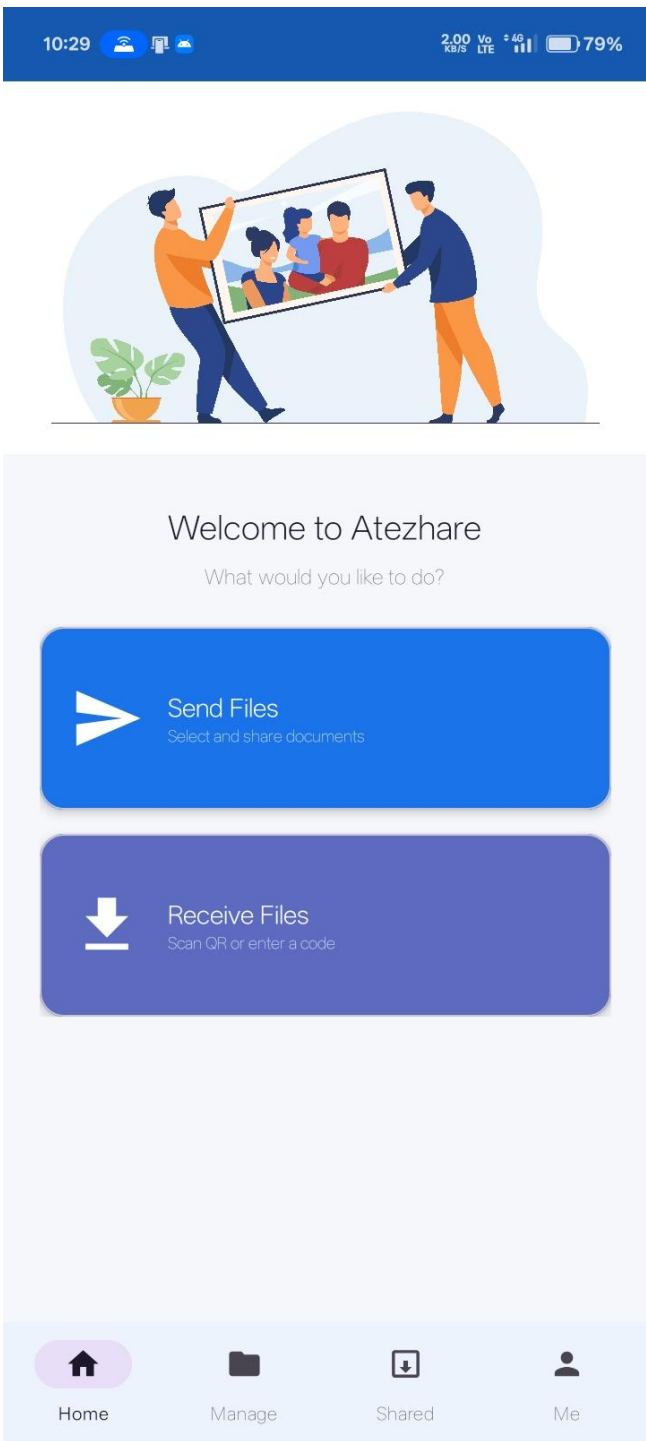


# Tools And Methodology Used

- **SpringBoot : Backend and api management**
- **Android :Main application interface**
- **Render : Backend hosting**
- **Supabase : Datamanagement**
- **Android Studio: Android app development**
- **Spring tools : Backend development**



# Simulation And Implementation



# References

- 1) William Stallings, Cryptography and Network Security,
- 2) Pearson Education.Andrew S. Tanenbaum, Computer Networks,
- 3) Pearson Education.Android Developers Official Documentation – WiFi Direct and Bluetooth APIs.OkHttp Official Documentation.
- 4) IEEE Research Papers on Secure File Sharing and Authentication.OWASP Mobile Security Guidelines.

**Thank you**